

CASE STUDY
PROFILE OF CLINICAL TRIAL
BIostatisticians AND UNIVERSITY
COURSES TO TRAIN THEM IN SOUTH AFRICA

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Summary: Many statistical issues in the area of clinical trials are specific to this particular field. A clinical trial biostatistician should not only be appropriately qualified in general statistical theory, but also be trained and experienced in the application of statistics to clinical trials. This study investigated the background and training of statisticians practising in this field in South Africa, and the training available to these biostatisticians at South African universities.

A literature study was conducted to determine the profile of a clinical trial biostatistician internationally, and a review of the content of statistics courses at South African universities was performed. A questionnaire survey was conducted to assess the education and training profile of such biostatisticians in South Africa and to assess the knowledge of biostatisticians in specific areas considered necessary when practising as a clinical trial biostatistician.

This paper presents the findings of this research and recommendations based on the findings.

1. Introduction

Clinical trials comprise a substantial set of related steps, starting with study design (including protocol writing, case report form (CRF) design and randomisation) and regulatory submission, continuing through the clinical conduct of the study, data management and statistical analysis, and culminating in the reporting of the study results. In recent years the importance of biostatistical input into these steps has been recognised, and such input has been made mandatory by regulatory agencies (International Conference on Harmonisation, 1996; 1998; Department of Health, 2000). This regulation has resulted in an increasing number of statisticians being employed in the pharmaceutical industry and by regulatory authorities where statisticians review applications for the marketing approval of medicinal products (Kopcke, Jones, Huitfeldt & Schmidt, 1998).

The *International Conference on Harmonisation (ICH)* E9 guidance on Statistical Principles for Clinical Trials (International Conference on Harmonisation, 1998) is specific regarding the requirements for the clinical trial biostatistician, "...it is assumed that the actual responsibility for all statistical work associated with clinical trials will lie with an appropriately qualified and experienced statistician, as indicated in ICH E6 (International Conference on Harmonisation, 1996). The role and responsibility of the trial statistician, in collaboration with other clinical trial professionals, is to ensure that statistical principles are applied appropriately in clinical trials supporting drug development. Thus, the statistician should have a combination of education/training and experience sufficient to implement the principles articulated in this guidance."

The European Federation of Statisticians in the Pharmaceutical Industry (EFSPI) working party agreed that a “qualified medical statistician” is expected to have a university degree in statistics, or equivalent, plus more than three years of experience in medical statistics (EFSPI Working Group, 1999). However, the working party conceded that this definition was vague since it lacked detail regarding the content of university courses.

Phillips (1999) specified the qualifications and experience required from statistical personnel in Phase II/III clinical research for different levels of responsibility, as outlined in Table 1.

Table 1 Qualification and experience for different levels of statistical personnel in Phase II/III clinical research

Title	Qualification	Experience (years)	Level of Responsibility
Statistician	BSc	–	Study
Senior Statistician	MSc, PhD	≥ 5 years	Project
Principal Statistician	MSc, PhD	≥ 10 years	Therapeutic Area

Chuang-Stein (1996) summarised the desirable traits of a clinical trial biostatistician and discussed the on-the-job training of a graduate statistician for developing these traits. It was suggested that, aside from sound training in statistical theory, a statistician should know the basic elements of both drug development and relevant therapeutic areas. Statisticians should know regulatory requirements for drug approval, remain up to date with the computer facilities relevant to their area of support, and develop the skills required for the effective communication of statistical concepts and ramifications. These traits are similar to the skills of an effective industrial statistician as listed in a Section on Statistical Education of an American Statistical Association (ASA) report by the Committee on Training of Statisticians for Industry Preparing Statisticians for Careers in Industry (1980).

The clinical trial industry in South Africa has grown substantially over recent years, with an estimated growth of 40% from 1997 to 1998 (Christley, 1998). This growth has continued with an estimated yield of R826 million in 2000 (Department of Health, 2000). Together with the growth of this industry, it can be expected that there has been an increase in the number of statisticians working as clinical trial biostatisticians in South Africa.

2. Aim of the Study

An understanding of the profile (education/training, knowledge, years of experience, part of the industry in which they are employed) of clinical trial biostatisticians internationally, and how biostatisticians in South Africa compare with this profile, will yield many benefits. It will enable South African employers to recruit and train new biostatisticians effectively and to develop their existing biostatisticians in any areas where training needs may be identified. Individual biostatisticians will be able to identify topics in which to undergo further training and development, if necessary.

An assessment of the training needs of clinical trial biostatisticians in South Africa, and how these needs compare to what is offered at South African universities, will enable the universities to identify how they can better serve medical research and the pharmaceutical industry. This research can also contribute to the development of a comprehensive manual or training programme to accommodate the learning needs of a clinical trial biostatistician and may subsequently be used in either a university or business context.

3. Methods

Literature study

A literature study was performed to investigate how literature defines an “appropriately qualified and experienced biostatistician”. Furthermore, literature was consulted regarding which training/education issues are important for a clinical trial biostatistician, to ensure that these topics were included in the questionnaire survey. The literature sources included journals from statistical, medical and pharmaceutical fields of study, books on medical statistics and clinical trial methodology, applicable national and international guidelines.

Survey of clinical trial biostatisticians

A questionnaire survey was designed to investigate the profile of clinical trial biostatisticians in South Africa. The questionnaire was designed to assess the knowledge of respondents in areas about which appropriately educated/trained clinical trial biostatisticians should have knowledge and to collect information on how they acquired this knowledge (university degree, on-the-job training, formal corporate training, self-study/reading).

The sampling frame for this questionnaire survey intended to include all clinical trial biostatisticians in South Africa. For the purpose of this study a clinical trial biostatistician referred to a statistician responsible for providing statistical input at any stage of a clinical trial, from the design through to the final study report, who analysed at least one clinical trial per year.

To establish a sampling frame various institutions were approached that might employ or come into contact with clinical trial biostatisticians. The institutions consisted of pharmaceutical companies, university research or statistics/mathematical statistics departments, university medical faculties or

pharmacy departments, contract research organisations (CROs), regulatory authorities and other governmental research organisations. In order to include clinical trial biostatisticians that possibly consult privately, individuals on the South African Statistical Association (2003) (SASA) Consultant List were approached. Due to the small number of clinical trial biostatisticians employed in South Africa the entire population, as identified in the sampling frame, was surveyed.

The questionnaire was distributed by e-mail and it could either have been completed electronically and returned by e-mail or printed out, completed by hand and returned by regular mail. The questionnaire was tested and updated prior to distribution.

The data was summarised descriptively in contingency tables with absolute and relative frequencies. The results were presented overall and by industry in which the respondent was employed.

Survey of South African universities

A survey was conducted to establish what undergraduate statistical and biostatistical training is offered at universities in South Africa.

All universities ($n = 20$) with statistics/mathematical statistics or biostatistics departments in South Africa were surveyed. All other tertiary education facilities were omitted since the preliminary literature search indicated that a clinical trial biostatistician requires at least a university qualification in statistics or a related field (EFSPI Working Group, 1999).

Universities were requested to send copies of the curricula of the undergraduate courses offered by the departments. If there was a lack of response by the department a reminder with contact information to answer general questions about the survey was sent to the department by e-mail. The departments were provided with both stamped pre-addressed return envelopes

and electronic contact details to allow them to submit the information with ease. Universities were followed up until response or to a maximum of four times.

Various definitions of the term “course” are in use by the different South African universities. University courses range from three-month courses to full-year courses, but all courses are restricted to a specific year of study. Courses were counted as defined by each university and not normalised to be courses of equivalent length.

Each course description was compared to the topics included in the clinical trial biostatistician questionnaire and the topics that each course covered were recorded. This sorting was done independently by two statisticians with experience of working as clinical trial biostatisticians, one of whom also had experience as a statistics lecturer. Discrepancies were discussed and consensus reached.

The data was summarised descriptively using contingency tables with absolute and relative frequencies.

Ethics

The research proposal along with a copy of the clinical trial biostatistician questionnaire and the letters to be sent to identify clinical trial biostatisticians, to request information from the universities and to inform clinical trial biostatisticians about the survey, were submitted to the University of the Free State’s Ethics Committee with applicable supporting documentation. The Ethics Committee provided approval for the research to be conducted.

4. Results

Survey of clinical trial biostatisticians

Response rates

A total of 108 questionnaires were distributed to statisticians. The response per industry is summarised in the table below (Table 2):

Table 2 Response rates per industry

Industry	Not CTB	Delivery failure	Completed questionnaire	No response	Other	Total
CRO	0	0	11(100.0%)	0	0	11
University department	1(10.0%)	0	7(70.0%)	2(20.0%)	0	10
Government agency	5(29.4%)	0	8(47.1%)	2(11.8%)	2(11.8%)	17
SASA consultants list	12(18.5%)	35(53.8%)	3(4.6%)	9(13.8%)	5(7.7%)	65
Other	1(20.0%)	0	3(60.0%)	1(20.0%)	0	5
Total (%)	20(18.5%)	35(32.4%)	32(29.6%)	14(13.0%)	7(6.5%)	108

CRO = Contract Research Organisation

CTB = Clinical Trial Biostatistician

The response rate of institutions contacted to provide names and contact details for clinical trial biostatisticians was generally high. The response was also high for contacts who received their questionnaires, whether with a response that they had been incorrectly identified as a clinical trial biostatistician or a completed questionnaire. Aside from the SASA consultants list it appears that the sampling frame reasonably accurately represented the population being assessed. Many of the respondents working for government agencies who indicated that they did not practise as clinical trial biostatisticians stated that they worked as biostatisticians but were not exposed to clinical trials.

The respondents represent the population of South African clinical trial biostatisticians reasonably well with the exception of the statisticians employed by CROs. Since one of the two CROs that employ a relatively large number of biostatisticians refused to participate in the survey, most of the respondents from CROs originate from one company.

Profile of clinical trial biostatisticians

Qualifications

All the respondents have at least a bachelor's degree in statistics, with the majority (89.7%) having a post-graduate degree. More than half (62.1%) of the respondents have at least a master's degree in statistics.

Clinical trial biostatisticians employed by CROs mostly have honours degrees as their highest statistics qualification. In both university departments and government research agencies, nearly all respondents have either a master's degree or PhD as their highest statistics qualification.

Industry

Clinical trial biostatisticians are employed in establishments serving different parts of the pharmaceutical industry. The following table presents the industries in which respondents are currently and were previously employed (Table 3).

Table 3 Current employment by industry

Industry	n	%
Contract research organisation	10	34.5
University research/statistics department	9	31.0
Government research agency (MRC, Statistics SA)	8	27.6
Other, specify ^a	2	6.9
Total	29	100.0

^aThe other current industries that were specified are "University department and Private" and "NGO".

Experience: Years of experience

Over half of the respondents have less than five years of experience. There were three respondents (10.3%) in each of the categories of 6 to 10 years and 11 to 15 years. Almost a quarter of the respondents (7 respondents, 24.1%) have over 15 years of experience of working as a biostatistician.

Experience: Percentage of time as a clinical trial biostatistician

Only 8 respondents (27.6%) spent more than 80% of their time working as a clinical trial biostatistician. Another 8 respondents spent between 21% and 80% of their time in this role. Almost half of the respondents (13 respondents, 44.8%) spent less than 20% of their time working on clinical trials.

Experience: Number of trials analysed in a year

Twelve respondents (41.4%) analysed fewer than three clinical trials a year. Nine respondents (31%) analysed between four and six trials a year. Four respondents (13.8%) each analysed seven to ten and greater than ten trials, respectively.

Experience: Clinical trial phase

Respondents may have worked on clinical trials in more than one phase of development. Respondents mostly have experience in analysing Phase III clinical trials and have the least experience analysing Phase IV clinical trials. Sixteen respondents (55.2%) had each worked on Phase I and Phase II trials. Eighteen respondents (62.1%) had worked on Phase III trials and 11 respondents (37.9%) had worked on Phase IV trials.

Experience: Task experience

Biostatistical input into clinical trials is required in study design (including protocol writing, CRF design and randomisation), regulatory submission, the clinical conduct of the study, data management and statistical analysis, and

in reporting the study results. The experience of respondents in these various tasks is summarized in Table 4.

Table 4 Task experience

Task experience	n	%
	(N = 29)	
Writing clinical development plans	1	3.4
Writing clinical trial protocols	14	48.3
Designing case report forms	12	41.4
Statistical analysis of a clinical trial	28	96.6
Writing clinical trial reports	20	69.0
Sample size calculation	22	75.9
Preparation of randomisation materials	13	44.8
Preparation/review of data management plans	16	55.2

Fewer respondents based at university departments had experience in writing clinical trial reports than those at CROs or government research agencies. Clinical trial biostatisticians employed by CROs had less experience in sample size calculation but more experience in the preparation or review of data management plans than respondents in university departments or government research agencies.

Knowledge of clinical trial biostatisticians

Table 5 includes a list of all the topics that the questionnaire assessed with respect to the knowledge of respondents and how they acquired this knowledge. The response categories for the knowledge of respondents in each category were: no knowledge/experience, little knowledge/experience, theoretical knowledge only, working knowledge and extensive experience. The method by which respondents acquired knowledge was assessed in the following categories: university degree, on-the-job training, formal corporate training, self-study/reading and other training courses.

Table 5 Topics included in the Biostatistician questionnaire

Topic	No know/exp		Little know/exp		Theoretical know only		Working knowledge		Extensive experience	
	n	%	n	%	n	%	n	%	n	%
The drug development process and clinical trials										
History and rationale of RCT	3	10.3	1	3.4	6	20.7	15	51.7	4	13.8
Drug development process	9	31.0	5	17.2	9	31.0	5	17.2	1	3.4
Conduct of a clinical trial	2	7.1	2	7.1	3	10.7	19	67.9	2	7.1
Clinical trial team	3	10.3	7	24.1	4	13.8	12	41.4	3	10.3
Regulatory requirements and international guidelines										
ICH E3 – Structure and content of clinical study reports	11	37.9	4	13.8	1	3.4	10	34.5	3	10.3
ICH E6 – Good clinical practice	11	37.9	3	10.3	4	13.8	8	27.6	3	10.3
ICH E9 – Statistical principles for CTs	6	20.7	1	3.4	3	10.3	13	44.8	6	20.7
ICH E10 – Choice of control group in CTs	7	24.1	5	17.2	3	10.3	10	34.5	4	13.8
PSI SOP guidance documents	17	58.6	4	13.8	3	10.3	4	13.8	1	3.4
21 CFR Part 11 – Electronic records and electronic signatures	18	62.1	4	13.8	3	10.3	4	13.8		
Regulatory submission process	18	62.1	7	24.1	3	10.3			1	3.4
Guidelines for Good Practice in the Conduct of CTs in Human Participants in SA	14	48.3	8	27.6	3	10.3	3	10.3	1	3.4
IECs and IRBs	10	34.5	8	27.6	3	10.3	5	17.2	3	10.3
Clinical trial design										
Types of clinical trials (superiority, etc.)	4	13.8	4	13.8	3	10.3	15	51.7	3	10.3
Clinical trial designs (parallel, etc.)	1	3.4	2	6.9	5	17.2	14	48.3	7	24.1
Types of outcome variables (composite variables, etc.)	3	10.3	2	6.9	2	6.9	18	62.1	4	13.8
Sample size calculation	2	6.9	2	6.9	8	27.6	11	37.9	6	20.7
Statistical considerations for protocol review (including protocol amendments)	5	17.2	3	10.3	2	6.9	14	48.3	5	17.2
Randomisation and blinding	3	10.3	3	10.3	7	24.1	10	34.5	6	20.7
Statistical considerations for CRF design	7	24.1	4	13.8	5	17.2	6	20.7	7	24.1
Statistical considerations for interim analyses	5	17.2	5	17.2	5	17.2	10	34.5	4	13.8
Statistical considerations for DSMBs	16	55.2	3	10.3	7	24.1	3	10.3		
Avoiding bias in clinical trials	2	7.1	5	17.9	5	17.9	12	42.9	4	14.3
Data management										
Data management for clinical trials	2	6.9	4	13.8	4	13.8	15	51.7	4	13.8
Biostatistical input into data management processes	2	6.9	3	10.3	2	6.9	18	62.1	4	13.8
Conducting the statistical review of a database	6	20.7	4	13.8	3	10.3	10	34.5	6	20.7
Statistical analysis considerations										
Writing statistical analysis plans	3	10.3	3	10.3	2	6.9	12	41.4	9	31.0
Writing derived dataset specifications	9	31.0	7	24.1	3	10.3	8	27.6	2	6.9
Identifying analysis populations	6	20.7	2	6.9	1	3.4	15	51.7	5	17.2
Conducting a blind review of data	9	31.0	5	17.2			11	37.9	4	13.8
Presentation/analysis of demographic and background data	2	6.9					15	51.7	12	41.4
Presentation/analysis of laboratory data	2	6.9	2	6.9			19	65.5	6	20.7
Presentation/analysis of adverse events	5	17.2	3	10.3			16	55.2	5	17.2
Presentation/analysis of concomitant illnesses and medication, medical history and previous medications	5	17.2	2	6.9			17	58.6	5	17.2
Data analysis considerations (covariates, interactions, etc.)	3	10.3			1	3.4	18	62.1	7	24.1
International medical coding dictionaries	10	34.5	5	17.2			13	44.8	1	3.4
Programming considerations and documentation (Good programming practices)	4	13.8	3	10.3			17	58.6	5	17.2
Pharmacokinetics & Pharmacodynamics	10	34.5	2	6.9	5	17.2	9	31.0	3	10.3
Bioequivalence studies	9	31.0	6	20.7	3	10.3	6	20.7	5	17.2

Topic	No know/exp		Little know/exp		Theoretical know only		Working knowledge		Extensive experience	
	n	%	n	%	n	%	n	%	n	%
Statistical methods										
ANOVA/ANCOVA			1	3.4	2	6.9	10	34.5	16	55.2
Linear regression	1	3.4	1	3.4	4	13.8	11	37.9	12	41.4
Repeated measures/Multivariate analysis	1	3.4			5	17.2	12	41.4	11	37.9
Non-parametric methods	1	3.4			3	10.3	14	48.3	11	37.9
Contingency tables	1	3.4			2	6.9	11	37.9	15	51.7
Risk ratios / odds ratios / Mantel-Haenszel method	1	3.4	1	3.4	4	13.8	12	41.4	11	37.9
Logistic regression	2	6.9	2	6.9	4	13.8	12	41.4	9	31.0
Generalised linear models	1	3.4	1	3.4	3	10.3	13	44.8	11	37.9
Exact tests			2	6.9	3	10.3	12	41.4	12	41.4
Analysis of survival data (incl. Cox regression, Kaplan-Meier estimates)	3	10.3	3	10.3	3	10.3	12	41.4	8	27.6
Reporting										
Medical writing (scientific writing) style and conventions	3	10.3	7	24.1			11	37.9	8	27.6
Writing/reviewing publications	5	17.2	6	20.7	1	3.4	10	34.5	7	24.1
Writing/reviewing clinical study report	4	13.8	6	20.7			13	44.8	6	20.7
Quality control and documentation										
Quality control and review processes	7	24.1	4	13.8	3	10.3	12	41.4	3	10.3
Maintaining project documentation (including version control etc)	6	20.7	6	20.7	2	6.9	12	41.4	3	10.3
Paper and electronic archiving	7	24.1	5	17.2	1	3.4	15	51.7	1	3.4
Computer skills/packages										
MS Word							12	41.4	17	58.6
MS Excel							14	48.3	15	51.7
MS PowerPoint	1	3.4	3	10.3			17	58.6	8	27.6
Nqasy	20	69.0	4	13.8	2	6.9	2	6.9	1	3.4
StatOxact	20	69.0	3	10.3	1	3.4	3	10.3	2	6.9
WinNonLin	25	86.2	1	3.4	1	3.4	1	3.4	1	3.4
SAS			1	3.4			14	48.3	14	48.3
CTA	21	72.4	2	6.9			4	13.8	2	6.9

Note: If a respondent did not indicate their level of knowledge/experience this omission was taken as 'No knowledge/ experience'

Know = Knowledge, Exp = Experience, RCT = Randomised clinical trial, IECs = Independent Ethics Committees, IRBs = Institutional Review Boards, CTs = Clinical trials, SA = South Africa

The questionnaire sections in which respondents reported a good knowledge and experience were the following:

- The drug development process and clinical trials
- Clinical trial design
- Data management
- Statistical methods
- Computer skill and packages

The questionnaire sections in which respondents were less familiar were the following:

- Regulatory requirements and international guidelines
- Statistical analysis considerations
- Reporting
- Quality control and documentation

Overall knowledge and experience were mostly acquired through on-the-job training followed by self-study and reading. Thereafter the most frequent method of learning was formal in-house training with a few respondents indicating that they acquired their knowledge through a university degree. Very few respondents acquired their knowledge and experience through other training courses.

University course content survey

Response rates

Information was obtained for fourteen (70.0%) of the 20 South African universities.

Content of courses

Table 6 summarises the number of South African universities that have at least one course (first-, second- or third-year course) addressing the topics identified on the clinical trial biostatistician questionnaire.

Very few South African universities presented courses covering topics outside of the Statistical Methods Section as identified on the clinical trial biostatistician questionnaire. The most frequently covered courses other than those under statistical methods were

- courses identified as having a computer usage component,
- courses covering quality control (4 universities covered third-year courses in this topic), and
- projects which require students to collect, analyse and report data.

The following statistical methods topics are well covered by courses from first year through to third year:

Table 6 South African universities with at least one course addressing a specific questionnaire topic

Section	Topic	1st-year course		2nd-year course		3rd-year course		Any year	
		n	%	n	%	N	%	n	%
Clinical trial	Clinical trial designs	1	7.1	0	0.0	0	0.0	1	7.1
Design	Sample size calculation	2	14.3	1	7.1	0	0.0	3	21.4
	Randomisation and blinding	0	0.0	2	14.3	0	0.0	2	14.3
Computer	MS Word	1	7.1	0	0.0	0	0.0	1	7.1
skills/packages	MS Excel	2	14.3	2	14.3	1	7.1	4	28.6
	SAS	1	7.1	1	7.1	0	0.0	2	14.3
	Computer usage: package not specified	6	42.9	6	42.9	5	35.7	10	71.4
	Computer usage: package specified but not in questionnaire	3	21.4	0	0.0	3	21.4	5	35.7
Other	Project	1	7.1	2	14.3	3	21.4	5	35.7
Quality control and documentation	Quality control and review processes	1	7.1	0	0.0	4	28.6	5	35.7
Statistical methods	Analysis of continuous data	14	100.0	11	78.6	6	42.9	14	100.0
	ANOVA/ANCOVA	11	78.6	8	57.1	9	64.3	13	92.9
	Linear regression	13	92.9	9	64.3	8	57.1	13	92.9
	Repeated measures/Multivariate analysis	1	7.1	6	42.9	6	42.9	10	71.4
	Non-parametric methods	9	64.3	3	21.4	5	35.7	12	85.7
	Analysis of binary and categorical data	14	100.0	9	64.3	3	21.4	14	100.0
	Contingency tables	7	50.0	3	21.4	2	14.3	9	64.3
	Risk ratios / odds ratios / Mantel-Haenszel method	1	7.1	0	0.0	1	7.1	2	14.3
	Logistic regression	1	7.1	3	21.4	7	50.0	8	57.1
	Generalised linear models	0	0.0	0	0.0	1	7.1	10	71.4
	Exact tests	0	0.0	0	0.0	0	0.0	0	0.0
	Analysis of survival data (incl. Cox regression, Kaplan-Meier estimates)	1	7.1	0	0.0	5	35.7	5	35.7

Note: A university course may differ in length, varying from a three month course to a twelve month course

- Analysis of continuous data (which includes the comparison of continuous data, p-values, point estimates and confidence intervals),
- ANOVA/ANCOVA,
- Linear regression,
- Analysis of binary and categorical data (which includes the comparison of binary and categorical data, p-values, point estimates and confidence intervals),
- Generalised linear models (at a third-year course level),
- Non-parametric methods (to a lesser degree than the previous courses, but still reasonably well covered).

Statistical methods topics that are included in courses from about half of the South African universities that responded to the request for information are

- Repeated measures/multivariate analysis,
- Contingency tables,

- Logistic regression.
- Analysis of survival data including Cox regression and Kaplan-Meier estimates was included in 35.7% of the third-year courses at the South African universities.

Statistical methods topics as identified on the clinical trial biostatistician questionnaire which are not well covered by courses presented by South African universities are exact tests, risk ratios, odds ratios and Mantel-Haenzel methods.

5. Discussion

Survey of clinical trial biostatisticians

Profile of clinical trial biostatisticians

Qualifications

The minimum industry qualification requirement for clinical trial biostatisticians in Europe (EFSPI Working Group, 1999) is at least a three-year degree in statistics or an equivalent qualification. More than half of the respondents have at least a master's degree in statistics implying that in respect of qualifications, these respondents qualify to act as senior or principal statisticians as defined by Phillips (1999).

Industry

The type of industry in which clinical trial biostatisticians are employed in South Africa differs from the profile seen in Europe. Since pharmaceutical companies do not have research offices performing statistical analyses in South Africa they are not a potential employer of clinical trial biostatisticians. CROs seem to employ most of the clinical trial biostatisticians.

Experience: Years of experience

The criteria defined by Phillips (1999) (Table 1) to indicate the acceptable levels of responsibility for clinical trial biostatisticians included the years of relevant experience. A principal statistician should have more than ten years of relevant experience and a senior statistician should have more than five years of relevant experience.

Over half (55.2%) of the respondents have less than five years of experience and this fact may be indicative of the recent growth of this industry in South Africa and the increased acknowledgement of the necessity for biostatistical input into clinical trials. This lack of experience was observed in all three industry subgroups. Half of the respondents would be able to act as either senior or principal statisticians based on their level of experience.

Experience: Percentage of time as a clinical trial biostatistician

Respondents employed by CROs spend a higher percentage of their time working as clinical trial biostatisticians than those employed in government agencies and university departments. The difference observed in the percentage of time spent as a clinical trial biostatistician is as expected because in CROs a biostatistician's primary line of business is being involved in the design, analysis and reporting of clinical trials. In contrast, university biostatisticians are potentially involved in a range of research being performed at a university, and clinical trials usually form a small part of the research being performed at these institutions. Biostatisticians working for government agencies are involved in a small number of clinical trials with most of their time being used for epidemiological studies.

Experience: Number of trials analysed in a year

The number of trials analysed in a year is reasonably well related to the percentage of time spent working as a clinical trial biostatistician. The profile

of number of trials analysed in a year is as expected, given that many of the clinical trial biostatisticians employed by universities are involved in research other than clinical trials. Also the government research agency clinical trial biostatisticians are typically involved in more epidemiological work. CRO clinical trial biostatisticians are specifically appointed to analyse clinical trials and therefore they analyse a higher number of trials in a year. This contrast may possibly imply that CRO clinical trial biostatisticians are likely to have a wider exposure to clinical trials.

Experience: Clinical trial phase

Clinical trial biostatisticians may have experience in analysing clinical trials in different phases of drug development. Clinical trials conducted during the different phases of development often have different design and analysis requirements. It is preferable that statisticians with experience in a particular phase of the development process should be involved in the design and analysis of trials in that development phase. It would be advisable that prior to becoming involved in studies in phases of development with which the clinical trial biostatistician is not familiar, the clinical trial biostatistician should learn about issues specific to that phase.

Task experience

Mostly the principal statistician from a pharmaceutical company is involved in writing clinical development plans. Since none of the pharmaceutical companies in South Africa employ clinical trial biostatisticians there is a lack of experience in this task. Approximately half of the respondents had experience in most of the tasks involved in clinical trial design, usually performed by senior biostatisticians. All clinical trial biostatisticians should at least have a level of responsibility comprising the statistical analysis of a clinical trial and the co-authorship of clinical trial reports (Phillips, 1999;

Cook, 1995). South African biostatisticians mostly had experience in these areas of trials.

Knowledge of clinical trial biostatisticians

The areas in which respondents were most familiar were those topics appropriate to statistical consulting in general as well as the overview of drug development and clinical trials. The latter is as could be expected from a group responsible for the analysis of clinical trials.

The questionnaire sections in which respondents were least familiar, were the more advanced topics which are directly related to the analysis of clinical trials. This can be understood in the context of many of the respondents not being uniquely responsible for the analysis of clinical trials but also other types of analyses, thus not having the leeway to concentrate exclusively on topics appropriate to clinical trials. Many of the topics in which they were least familiar, were with respect to international guidelines and requirements.

University course content survey

The courses offered by responding universities include basic statistical methodology and computer skills. However, research methodology courses, courses covering clinical trial and experimental design and other statistical analysis considerations are not offered. There were also no courses specific to statistics for medical research. Epidemiology is primarily offered to health professionals in South Africa with such courses being taught in medical schools.

According to Zelen (2003) biostatisticians should be broadly trained in statistics, probability, computing and applied mathematics. They should be familiar with a broad range of methodology, must have a sound knowledge of theory and must be capable data analysts.

Although it is not necessarily appropriate for statistics departments at South African universities to train students in all of the topics included in the questionnaire, given the small population of clinical trial biostatisticians in South Africa, they should at least provide students with the broad knowledge of the topics listed in the statistical methods section. Many of the topics appropriate for clinical trial biostatisticians are also of relevance to biostatisticians in general. Altman's book "Practical Statistics in Medical Research" (Altman, 1991) covers most of the relevant methods and could thus be used as a general textbook in a biostatistics course.

The statistical methods, which should be considered for inclusion in undergraduate courses are

- Exact tests
- Risk ratios/odds ratios/Mantel-Haenszel methods
- Analysis of survival data
- Repeated measures/multivariate analysis
- Contingency tables
- Logistic regression

Alternatively, it may be more viable to develop an honours or master's Biostatistics programme to be taken by students completing a general statistics undergraduate programme. Some South African universities already offer such a course, for example, the University of Cape Town has a Biostatistics honours course and the University of Fort Hare offers a Biostatistics master's programme. The University of Stellenbosch has recently developed a post-graduate Biostatistics programme.

If a postgraduate programme is developed by a South African university it would be worthwhile to investigate the content of such programmes offered overseas. Tobi *et al.* (2001) propose six disciplines for a curriculum for

the consultation biostatistician: applied statistics, methodology, epidemiology, communication, computational science and personal effectiveness.

The following courses can be considered for inclusion in a post-graduate biostatistics programme:

- Biostatistics and epidemiology, including statistical methods appropriate to biostatisticians
- Techniques for (bio)statistical consulting, including non-technical skills, for example communication with clients, presentation skills and publication and report writing
- Clinical trials: Design of medical studies, with emphasis on randomized controlled clinical trials; analysis considerations for clinical trials, including topics such as determining analysis populations
- Research methodology: Good clinical practice, ethics, including consent and insurance, adverse drug reaction surveillance, randomisation, data management, protocol preparation, case record forms and international guidelines for clinical research

In order to facilitate the development of such a university programme it is recommended that an industry representative (at least one specific to the clinical trial industry) be included on the Standards Generating Body for Statistics. SASA has progressed in the different aspects required for such a body (De Wet, 2003) and has requested feedback from industry regarding their requirements for statistics courses.

Zelen (2003) also mentions that in training biostatisticians, students should be exposed to the application of biostatistics to real problems. Only five of the 14 universities participating in the survey offer a “project” course which involves the collection, analysis, interpretation and reporting of data. The implementation of such a course at all universities offering a statistics programme would benefit the students in making the transition from the academic environment to the workplace. Such a course can also familiarise students with basic research methodology, including how to write a protocol.

It would also be preferable if these skills could be linked to a course in experimental design.

In order to develop the computing skills of students, universities should promote the use of computers and preferably SAS[®] for assignments and the practical component of the courses. Tobi *et al.* (2001) propose computational science as one of the disciplines in which biostatisticians should be trained. They define computational science as the critical use of standard statistical software, including the writing of custom macros and possibly developing computer software.

Development of a clinical trial biostatistics manual

Since clinical trial biostatisticians in South Africa are predominantly trained through on-the-job training, and self-study and reading, it is recommended that a comprehensive manual be developed to be used when conducting such training. On-the-job training and self-study both have limitations as a method of training because there is no control over the content and completeness of such training. A manual can be developed and reviewed by qualified and experienced clinical trial biostatisticians and thus overcome these limitations.

Should such a manual be developed, it is recommended that it be updated regularly, perhaps yearly, since many of the guidelines mentioned in the “International Guidelines and Standard Operating Procedures” section are being constantly updated and new guidelines are being written.

Certification of medical statisticians in South Africa

The EFSPI Working Group (1999) discusses the development of a European certification scheme. However, due to countries not being able to find consensus with respect to certification or the profile of such a biostatistician, differing educational systems in Europe and possible legal implications, the

implementation of such a scheme is not feasible for the short term. However, it is hoped that the European-wide definition of an appropriately qualified and experienced statistician being one with a three-year degree in statistics and at least three years of experience in medical statistics, would make the implementation of such a scheme possible in the long-term.

Certification schemes are in place in the United Kingdom, Germany and Spain. The Netherlands is in the process of developing a certification scheme (EFSPI Working Group, 1999).

It is recommended that a certification scheme be developed and implemented in South Africa. Such a scheme should focus on the qualifications and experience of individuals. It should also specify the areas in which an applicant should be proficient. The implementation of such a scheme would promote confidence of employers and non-South Africans in the abilities of certified individuals. This certification should be an optional process by which individuals can achieve professional recognition for statistical knowledge and achievement. It should also be distinguished from the legal licensing of practitioners.

It should also be investigated if it would be possible to link up the certification of South African clinical trial biostatisticians with one of the certification programmes available in Germany, Spain and the United Kingdom.

6. Conclusion

The profile of clinical trial biostatisticians in South Africa, with respect to qualifications and experience, is comparable to clinical trial biostatisticians in Europe. However, the industries in which the biostatisticians in South Africa

are employed, differ from those that employ clinical trial biostatisticians in Europe.

South African clinical trial biostatisticians are not necessarily familiar with all the topics applicable to their discipline. The areas in which they were the least familiar were: regulatory requirements and international guidelines, statistical analysis considerations, reporting, and quality control and documentation. Aside from statistical methods which were mostly learned through a university degree, knowledge and experience were mostly acquired through on-the-job training followed by self-study and reading.

It is hoped that the implementation of a university programme specific to clinical trial biostatisticians, improvements in current statistical courses, the development of a clinical trial biostatistician manual and the introduction of a medical statistician certification scheme, would contribute to developing what Iman (1995) is referring to when he quotes Kettenring in saying, "Industry needs holistic statisticians who are nimble problem solvers".

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